Propeller Ceiling Effect in Forward Flight

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The study extends the investigation of the fixed-pitch small-scale propeller in the ceiling effect to forward flight conditions at different propeller incidence angles. Force-based experiments, phase-locked particle image velocimetry (PIV), and surface oil flow visualization were conducted on two APC propellers at the University of Dayton Low Speed Wind Tunnel. For propellers in edgewise flight, the power required at constant thrust decreases at small advance ratios for each h/D and then increases with a further increase in the advance ratio. Tilting the propeller forward reduces the ceiling effect benefits in both thrust and power, particularly at higher advance ratios. Performance similarity in the propeller ceiling effect at different h/D is observed, and a performance prediction method is proposed. Phase-locked PIV showed an increase in the propeller inflow angle in the ceiling effect at small advance ratios, resulting in higher thrust generation. This effect reduces with an increase in the advance ratio due to minimized interactions with the ceiling plate. Hence, the measured propeller in-ceiling-effect (ICE) propeller performance cannot represent the propeller ICE performance at a higher forward flight speed. Additionally, PIV and surface flow visualization indicated the presence of a stagnation point on the ceiling plate near the trailing side of the propeller disk at higher advance ratios, leading to a reduction in propeller thrust generation.

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Nomenclature

C_P	=	power coefficient
C_Q	=	propeller torque coefficient
C_T	=	propeller thrust coefficient
c	=	propeller chord, m
D	=	propeller diameter, m
h	=	distance to the ground/ceiling plate, m
J	=	advance ratio
J^*	=	normalized advance ratio
J_S	=	scaled advance ratio
n	=	propeller rotational speed per second
P	=	power, W
Q	=	torque, N · m
R	=	propeller radius, m
r	=	propeller radial location, m
T	=	thrust, N
V	=	flow velocity, m/S
V_i	=	propeller-induced velocity, m/S
V_w	=	propeller wake velocity, m/S
V_∞	=	freestream velocity, m/S
γ	=	propeller pitch, m
δ	=	propeller forward tile angle, deg
θ	=	propeller incidence angle, deg
ρ	=	air density, kg/m^3
σ	=	propeller solidity
ϕ	=	propeller local blade pitch, deg
φ	=	propeller inflow angle, deg
$\chi_{\text{eff}_{\text{IGE}}}$	=	effective propeller wake skew angle, deg
χ_{prop}	=	propeller wake skew angle, deg

Subscripts

ICE	=	in ceiling effect
OCE	=	out of ceiling effect
X, Y	=	inertial frame coordinates
x, y	=	body frame coordinates

I. Introduction

THE ground effect on lifting rotor/propeller performance during hovering has been studied theoretically [1] and experimentally [2–6] for decades. Placing the propulsor adjacent and parallel to a ground plate offers two benefits: an increase in thrust produced and a decrease in power required for a given rotational speed. However, the latter benefit is usually ignored in many small-scale propeller-driven UAV studies. Several ground effect models for hovering have been developed for large rotors, including Betz's ground effect model [1], Cheeseman's model developed based on Betz's theory [2], and Hayden's model developed based on experimental data [4].

The study of the rotor ground effect in forward flight has been extensively researched for larger rotors through analytical and experimental methods, including wind tunnel testing and flight tests [2,5–10]. The normalized advance ratio J^* introduced by Curtiss et al. [6] is widely used in studying the ground effect in forward flight. J^* represents the ratio of the freestream velocity V_∞ and the propeller-induced velocity V_i, which determines the overall characteristics of the rotor wake. At $0.45 < J^* < 0.7$ and within $h/D \leq 0.3$, an increase In Ground Effect (IGE) power consumption was observed for the rotor. At this range of advance ratios, a rotor leading-edge recirculation/vortical flow region was observed, resulting in a loss in rotor thrust generation at the same rotor rotational speed. Hence, higher power consumption is required to generate the same thrust [2,5,6]. With a further increase in the freestream velocity, between $0.7 < J^* < \sim 1.2$, the ground vortex was observed. The ground vortex disappears with further increments in J^* [6–8], although the exact J^* where the ground vortex disappears depends on the experimental conditions [8].

The ceiling effect, where the ceiling plane is placed on the inflow side of the rotor, has been rarely studied for large rotors due to practical limitations. However, with the increasing popularity of propeller-driven UAVs that can operate in confined spaces, the ceiling effect must be taken into account. Several recent studies have investigated the ceiling effect [11–17], including our previous work [18,19], which considers the propeller ceiling and the partial ceiling effect in hovering. Similar to the classical ground effect, the ceiling effect also results in thrust augmentation and power savings. A lower propeller pitch-to-diameter ratio, γ/D, results in a higher benefit in both ground and ceiling effects. The partial ceiling effect can also be modeled as a function of ceiling area, providing a straightforward method for evaluating the performance of hovering propellers in the ceiling effect. Sanchez-Cuevas et al. [11] studied the full and partial

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ground and ceiling effects on a 0.23 m propeller. An increment in thrust was found when flying above or below the obstacles, resulting in a significant pitch-up moment of the drone that caused unstable flight conditions. Conyers [13] studied the ground and ceiling effects of a quad-rotor with 0.23 m × 0.12 m propellers. Different from the ground effect, where IGE thrust augmentation is significantly affected by the rotor-to-rotor distance [13,20], the difference between the single-rotor and multirotor in-ceiling-effect (ICE) performance was relatively small. Carter et al. [14] and Hsiao and Chirarattananon [16] studied the ground and ceiling effects on a micro-quadrotor with a rotor diameter of 0.023 m. A similar ICE performance change of the micro-quadrotor was found when compared to a larger rotor studied by Sanchez-Cuevas et al. and Conyers, indicating good scalability of the ceiling effect. The particle image velocimetry (PIV) experiment conducted by Carter et al. [14] revealed a 22% reduction in the rotor downwash in the ceiling effect, which was much higher than in a traditional ground effect, which was approximately 6–10%. All four studies acknowledged the similarity between the ceiling and ground effects. However, the classical ground effect prediction model cannot be directly applied to the ceiling effect, as the ceiling effect occurs at a much smaller distance with a higher thrust augmentation when compared to the ground effect, which was also concluded in our previous study [18,19].

Taking advantage of the ceiling effect benefit, several ICE drone concepts are designed and studied by Jimenez-Cano et al. [15], Hsiao and Chirarattananon [16], and Nohara et al. [17]. These ICE drones can be used for close ceiling inspections or power-saving flights when constantly operating in the ceiling effect, with a potential power saving of up to 50% [16].

However, none of the aforementioned ceiling effect studies was conducted in forward flight conditions. The study conducted by Gao et al. [21] tested the feasibility of a quad-copter flying autonomously in close proximity to the ground and ceiling plate for power-saving purposes. A 4.5% saving in power was observed when flying in ground and ceiling effects in comparison to the out-of-ground and ceiling effect flight along the same flight path. However, the forward flight speed is relatively low ($J < 0.15$). Based on our previous study [22] and ground effect forward flight studies [2,5–10], the rotor ground effect performance changes dramatically in forward flight, especially at higher forward flight speeds. Hence, the current investigation extends the prior study to include propeller performance in the ceiling effect in forward flight at two incidence angles that provide insight into the effects of propeller geometry and incidence angles in forward flight near a ceiling plane, which can provide good design guidance for drones operating in the ceiling effect.

II. Experimental Setup

A. Force-Based Experiment

Experiments were conducted in the open jet configuration of the University of Dayton Low Speed Wind Tunnel (UD-LSWT). The UD-LSWT features a 16:1 contraction ratio, 6 antiturbulence screens, and a 0.762 m × 0.762 m open jet test section. To simulate the ceiling plate, a 0.914 m × 1.829 m aluminum alloy sheet with a thickness of 0.005 m was used in the experiment. The aluminum sheet is reinforced by six 80–20 aluminum extrusion beams at the bottom to ensure the rigidity of the sheet. All propellers were driven by an E-Flite Power 60–400Kv brushless out-runner electric motor controlled by a Hobbywing Platinum V3 electric speed controller (ESC) in governor mode to maintain a relatively constant rotation speed with a variation of less than 0.5%. The propeller and motor were powered by a PSW 30–108 constant-voltage power supply. The schematic of the experimental setup is illustrated in Fig. 1.

The propeller motor was attached to an ATI Industrial Automation Mini-40 six-component force balance, where the axial force along the z axis of the sensor is defined as thrust, and the rotational moment about the z axis is defined as torque. The propeller power coefficient was calculated directly from the measured propeller torque instead of the power supply output to eliminate the losses due to heat generation. The balance was bolted to an aluminum extrusion frame secured to the inlet of the plenum. The balance had a maximum

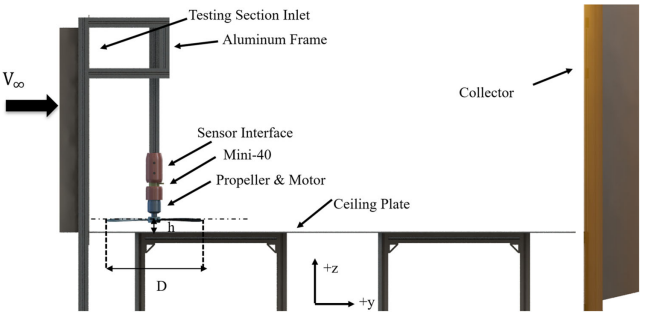


Fig. 1 Schematic of the experimental setup of the propeller ceiling effect test stand in UD-LSWT.

range of 120 N of force on its z axis and a maximum torque of 2 N · m about its z axis. The thrust and torque range encountered during the experiment were 5–25 N and 0.1–0.35 N·m, respectively. The uncertainty of the load balance is 0.75% on the x and y axes and 1% on the z axis, based on the manufacturer's calibration report. The air density of the experimental environment was calculated based on the temperature and elevation of the facility using the international standard atmosphere, resulting in an air density of 1.189 kg/m^3. The freestream velocity is measured by a TSI pitot tube mounted at the exit of the wind tunnel contraction and recorded by a TSI 5825 Micromanometer. The propeller rotational speed was measured using a 50 mW laser and a Thor Labs PDA10A photodiode. Based on the Kline–McClintock uncertainty calculations [23], the average aforementioned instrument uncertainty is calculated to be 1.22% based on thrust for the APC 14 × 6 propeller at its medium forward flight speed under 100 Rotation Per Second (RPS).

The data were sampled at a rate of 1000 Hz for a duration of 15 s at each data point. Two trials were conducted in each case to ensure data repeatability. To minimize sensor drift, tare values were taken before and after each test with the propeller powered off. The average of the two tares was subtracted from the respective thrust and torque readings. If the difference between the before and after tare values was greater than 0.1 N of force on any axis, the experimental data was rejected, and the experiments were run again. To filter out the vibration frequency caused by the propeller rotation and the natural frequency of the aluminum test frame, a finite impulse response (FIR) band-stop filter [24] was applied to the data. The tap tests on the test stand were performed with the propeller powered off, resulting in the structure's natural frequency of 60 Hz. The average random uncertainty in the force measurement based on the measured standard deviation is ±9.4% across all advanced ratios tested, while the average uncertainty in torque is ±7.2% after filtration. However, the results presented show excellent repeatability from both experimental runs.

Three propeller rotational speeds were tested to investigate the Reynolds number effect and the repeatability of the experiment. To achieve different advance ratios, the propeller rotational speed was kept constant, while the wind tunnel freestream velocity was changed accordingly, between 0 and 15 m/s. The propeller is mounted at a distance of 0.1524 m from the contraction exit. As the wind tunnel houses a 16:1-contraction-ratio inlet with a fifth-order polynomial cross section and a ratio of length to hydraulic diameter (L/D) of 1.0, the ratio of boundary-layer displacement thickness to the hydraulic radius of the test section inlet is much less than 1.0 [25]. In this case, considering a negligible boundary-layer thickness at the exit of the contraction, the boundary-layer growth on the ceiling plate between the testing section inlet and the propeller disk is 0.00272 m based on the Blasius laminar boundary-layer equations. When normalized by the propeller diameter, the boundary layer entering the propeller blade is approximately 0.765% of the APC 14 × 6 propeller diameter. Hence, the boundary-layer effects in the present study are considered to be negligible.

The propeller ceiling distance was measured from the center of the propeller hub to the ceiling plate. This distance was then normalized by the propeller diameter, resulting in the normalized ground distance, h/D. The propeller was mounted at a 90 deg incidence angle θ,

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Table 1 Force-based experiment testing matrix

Testing condition	Value	
Propeller	APC 14 × 6	APC 11 × 7
Diameter (D), m	0.3556	0.2794
Pitch (γ), m	0.1524	0.1778
Blade pitch angle at 75% radius (φ), °	10.3	15.1
Number of blades	2	
Solidity (σ)	0.0808	0.0849
Rotational speed (n), RPS	83.3, 100, and 116.7	
Freestream velocity (V_∞), m/s	0–15	
Advance ratio (J)	0–0.5	
Normalized ceiling distance (h/D)	−0.07 to −0.86	
Propeller incidence angle (θ), °	75–90	

where the propeller disk was parallel to the freestream, representing the low-speed hovering and edgewise flight conditions. A 75 deg incidence angle was also tested, which represents a nose-down forward flight condition.

Table 1 shows the testing matrix of the experiment. Two off-the-shelf, two-blade APC propellers were chosen for the experiment based on their hovering ICE performance in our previous study [18,19]. The APC 14 × 6 propeller represents a low γ/D propeller where the ICE performance benefit was significant in hover, while the APC 11 × 7 propeller represents a medium-high γ/D propeller where the ICE performance benefit was less than the APC 14 × 6 propeller. The propeller geometry, including the local pitch angle ϕ distribution and the local blade chord $c(r)$ distribution, is shown in Fig. 2. The magnitude of φ for the APC 11 × 7 propeller is almost double that of the APC 14 × 6 propeller at the same r/R, due to its higher γ/D ratio. For the chord distribution, the APC 14 × 6 propeller has a slightly higher $c(r)/R$ closer to the blade hub within $r/R < 0.4$. Note that $c(r)/R$ values for both propellers are almost identical when $r/R > 0.4$, as both propellers belong to the "thin electric" category based on APC's categorization.

B. Phase-Locked Particle Image Velocimetry

Streamwise, phase-locked PIV was conducted to investigate the flowfield between the propeller and the ceiling plate at various ground distances and advance ratios. The PIV setup is shown in Fig. 3. The tracer particles for the experiments were generated by a ViCount smoke seeder with glycerin oil. The Quantel Twins CFR 300, 200 MJ/pulse neodymium-doped yttrium aluminum garnet (Nd:YAG) frequency-doubled laser was aligned with a 60 deg fan angle Powell lens to illuminate the flowfield. A 76.2 mm × 304.8 mm glass plate was used to allow the laser to pass through the ceiling plate. To avoid reflections from the motor and the mounting interfaces, the laser sheet was placed 25 mm away from the centerline of the propeller disk. The Imperx 2021 CCD camera was used with a Canon EF 100 mm f/1.4 lens to provide a field of view (FOV) of 127 mm × 127 mm. To trigger the camera shutter, a 50 mW, 532 nm

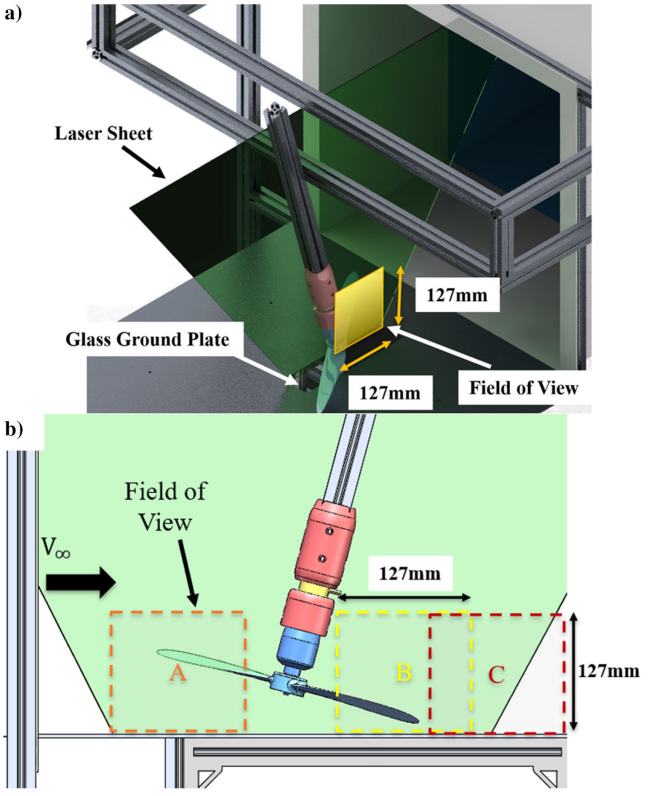


Fig. 3 Schematic of PIV experimental setup in UD-LSWT a) isometric view and b) side view.

laser, and a photodiode were used to trigger the Quantum Composers 9600 pulse generator, which captured the flow passing different propeller phases by changing the delay on the QC pulse generator.

The testing matrix for the PIV experiment is shown in Table 2. Not that 1000 image pairs were obtained per case and processed using the Fluere software (Version 1.3). Two iterations were performed in processing, with 64-pixel interrogation windows in the first iteration and 32-pixel interrogation windows in the second iteration. The time delay between the laser pulses was changed at each advance ratio to obtain 8–12-pixel particle displacement at the region of interest in the flowfield, including the propeller inflow at the FOV A and propeller wake at the FOV B and FOV C. According to the PIV uncertainty estimation developed by Lazar et al. [26], which includes uncertainty from equipment (instrumentation), particle dynamics, sampling, and image analysis (processing), the average uncertainty in velocity is around 3.59%. As the camera FOV is not large enough to capture the entire flowfield feature between the ceiling plate and the propeller disk in the ceiling effect, the camera's FOV was adjusted downstream, which is identified as the FOV A, FOV B, and FOV C in Fig. 3.

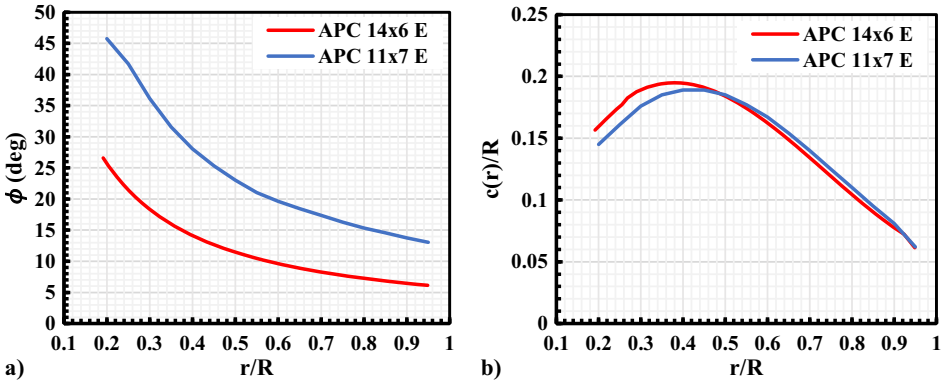


Fig. 2 a) Local blade pitch angle (ϕ) distribution and b) local blade chord ($c(r)$) distribution for APC 14 × 6 E and APC 11 × 7 E propeller.

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Table 2 PIV testing matrix

Testing condition	Value
Propeller	APC 14 × 6
Rotational speed (n)	90 RPS
Freestream velocity (V_∞)	0−15 m/s
Advance ratio	0.0, 0.1, 0.2, 0.3, 0.4
Normalized ceiling distance (h/D)	−0.2, −0.3
Propeller incidence angle (θ)	75°, 90°
Laser pulse frequency	∼15 Hz
Field of view	127 mm × 127 mm
No. of image pairs	1000/case

C. Smoke Visualization

Smoke flow visualization was conducted at the same condition as the PIV experiment to investigate the out-of-plane propeller inflow near the trailing side of the propeller disk. The experimental setup is shown in Fig. 4. The smoke wand has a diameter of 0.5 inches and is placed upstream of the propeller. A 300 mW continuous laser was used as a light source and paired with a 30-deg-angle Powell lens to create the laser sheet for illumination. The laser sheet is placed perpendicular to the freestream so that the out-of-plane propeller inflow, which cannot be captured by the PIV experiment (Fig. 3), can be captured by the smoke visualization, and the camera is placed downstream of the propeller.

D. Surface Oil Flow Visualization

The oil surface flow visualization was conducted to investigate the flowfield near the ground plate. A mixture of titanium dioxide, mineral oil, and oleic acid was applied to the ground plate coated with black enamel. The propeller was operating at 90 RPS and mounted at $\theta = 75°$ and a ground distance of 0.2 h/D, while the freestream velocity varies from 0 to 15 m/s to achieve $0 < J \leq 0.4$. The image was taken immediately after the propeller and wind tunnel were shut down.

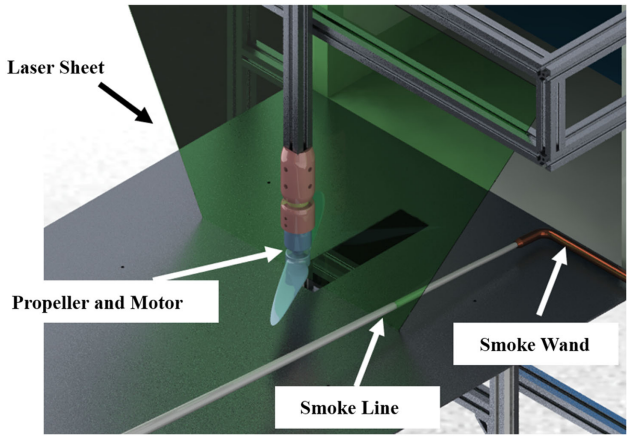


Fig. 4 Schematic of the smoke visualization experimental setup.

III. Results

A. Force-Based Experimental Results

As the change in propeller performance in the ceiling effect is reflected by the changes in thrust and power, both the force and torque of the propeller are measured. The force-based experimental results are shown in terms of normalized coefficients. Two different force and torque normalization methods are commonly used in the rotorcraft community; the first method is commonly used in the helicopter literature [Eq. (1)], while the other one is commonly used in the propeller literature [Eq. (2)] [27]. As the current study focuses on propeller performance, all coefficients are calculated based on the system used in propeller literature [27].

$$C_T = \frac{T}{0.5\rho(2\pi nr)^2\ \pi r^2} \quad C_Q = \frac{Q}{0.5\rho r(2\pi nr)^2 \pi r^2}$$
$$C_P = \frac{P}{0.5\rho(2\pi nr)^3 \pi r^2} \tag{1}$$

$$C_T = \frac{T}{\rho n^2 D^4} \quad C_Q = \frac{Q}{\rho n^2 D^5} \quad C_P = \frac{P}{\rho n^3 D^5} \tag{2}$$

1. *Results of* 14 × 6 *Propeller*

Since three different propeller rotational speeds (n) are tested in the current study, the Reynolds number effect on the propeller ICE performance can be investigated. An example of results for different rotational speeds of the APC 14 × 6 propeller at different distances to the ceiling plate (h/D) is shown in Fig. 5. The uncertainty of the force data measurement based on two standard deviations (95% confidence interval) is shown as the error bar in each plot. With the increment in n, the measured C_T and C_P at the same J and h/D shows a nonsignificant increase with the increase in Reynolds number. Therefore, it can be concluded that both APC propellers' performance in OCE and ICE is not sensitive to Reynolds number effects. As the overall difference in the measured C_T and C_p was less than 5% across the Reynolds number range tested, the experimental result from the 100 RPS case was chosen as a good representative to show the overall propeller performance in ceiling effect in forward flight.

Figure 6a shows the normalized thrust coefficient $C_{T_{\text{ICE}}}/C_{T_{\text{OCE}}}$ versus advance ratio J at different normalized ceiling distances h/D for the APC 14 × 6 propeller at $\theta = 90°$. The C_T of each datapoint was normalized by the C_T at out-of-ceiling-effect (OCE)

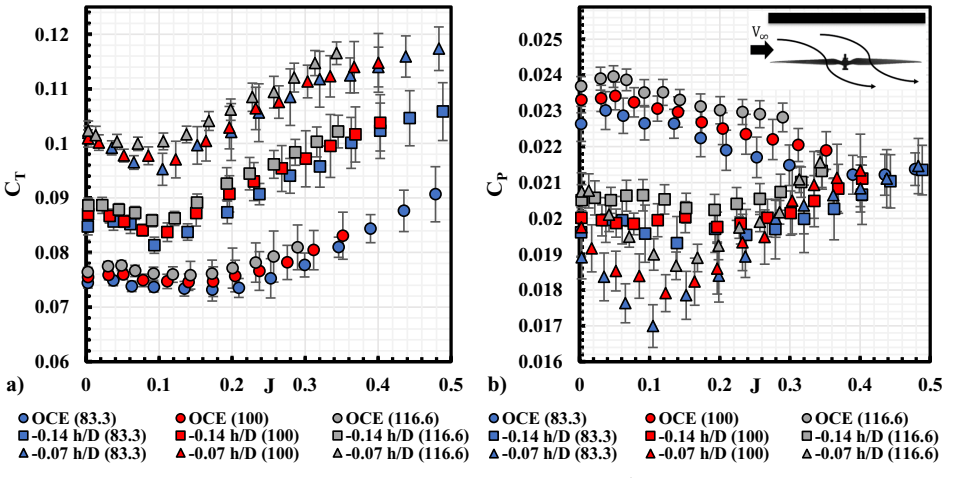


Fig. 5 Measured a) C_T and b) C_P at $\theta = 90°$ under various n and h/D in ceiling effect for APC 14 × 6.

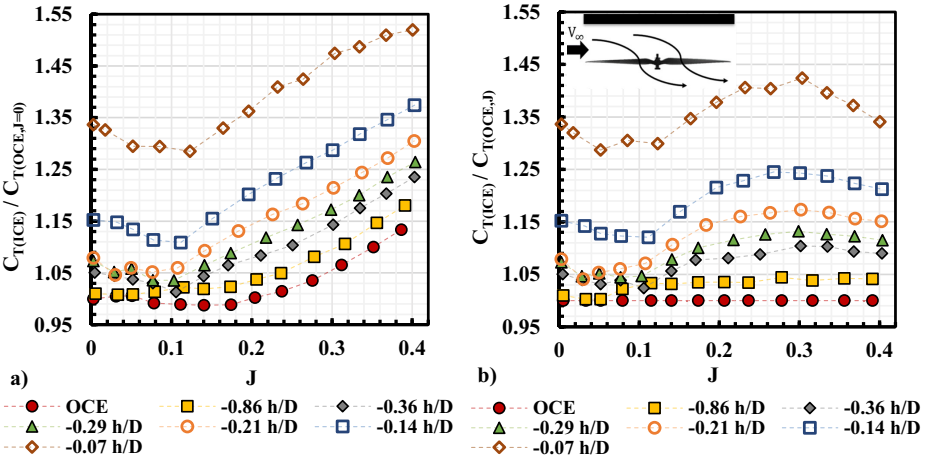


Fig. 6 Normalized thrust coefficient a) $C_{T_{ICE}}/C_{T_{OCE_{J=0}}}$ and b) $C_{T\ IGE}/C_{T\,OCE_J}$ at $\theta = 90°$ under 100 RPS in ceiling effect for APC 14×6.

hovering condition when $J = 0$, denoted by $C_{T_{OCE_{J=0}}}$. At $J < 0.1$, a clear reduction in C_T is observed with the increment of J for cases at $h/D < -0.36$, resulting in a loss in the thrust generation at a small forward flight speed. This is very similar to the results in a traditional ground effect study [5,6,10] and our previous ground effect study [22], where a loss in thrust is observed when the ground fountain vortex occurs at a low forward flight speed. The trend is then reversed at $J > 0.1$, where C_T continuous to increase with the increment in J. This differs significantly from the IGE cases in our previous study [22], where a further loss in C_T is observed in ground effect between $0.1 < J < 0.3$. For the results at different h/D, the difference between $-0.2 > h/D > -0.4$ is relatively small when compared to the cases at $h/D < -0.2$, where the increment in C_T becomes more dramatic at a small h/D, which is also observed in our previous study [18,19].

To eliminate the effect of freestream, an alternative normalization method is used since $C_{T\,OCE}$ also increases with the increment of J at $J > 0.1C_T$ of each h/D case is normalized by the C_T at OCE condition at the same J. This is shown in Fig. 6b. At $J < 0.1$, the overall trend is similar to Fig. 6a. At $0.1 < J < 0.3$, the increment in $C_{T_{IGE}}/C_{T_{OGE\,J}}$ is observed, however, at $J > 0.3$, $C_{T_{ICE}}/C_{T_{OCE\,J}}$ remains relatively constant at $h/D \geq 0.21$, indicating that the increment in C_T seen in Fig. 6a at the same J range is mainly a result of the increment in freestream velocity. At $h/D < 0.21$, a decrement in $C_{T_{ICE}}/C_{T_{OCE\,J}}$ is observed, indicating a reduction in the ceiling effect, which is also similar to the ground effect result in our previous study [22].

Figure 7 shows $C_{T\,ICE}/C_{T\,OCE}$ using both normalizations when tilting the rotor forward at $\theta = 75°$. For the OCE condition, $C_{T\,ICE}/C_{T\,OCE_{J=0}}$ decreases with the increment in J at $J < 0.2$ and then increases at $J > 0.2$. Again, the change observed in OCE is purely due to changes in freestream velocity. At $h/D > -0.21$, the overall trend for $C_{T\,ICE}/C_{T\,OCE_{J=0}}$ follows the OCE condition, with a higher $C_{T\,ICE}/C_{T\,OCE_{J=0}}$ observed for a smaller h/D. At $h/D < -0.21$, a significant reduction in C_T is observed at $J < 0.1$. C_T then increases with the increment in J. Again, when normalizing $C_{T\,ICE}$ by $C_{T\,OCE_J}$, the effect of the freestream is eliminated. In this case, for $h/D > -0.21$, $C_{T\,ICE}/C_{T\,OCE_J}$ increases slightly with the increment in J at $J < 0.1$, indicating a positive ICE benefit. $C_{T\,ICE}/C_{T\,OCE_J}$ remains relatively constant at $J > 0.1$, where the propeller performance is less affected by the ceiling proximity. While for $h/D = 0.16$, a small decrement in $C_{T\,ICE}/C_{T\,OCE_J}$ is observed at $J < 0.1$, $C_{T\,ICE}/C_{T\,OCE_J}$ then increases with the increment in J between $0.1 < J < 0.3$. Further increment in J results in a reduction in $C_{T\,ICE}/C_{T\,OCE_J}$, which is similar to the results for $\theta = 90°$ in Fig. 6. Moreover, the overall magnitude for $C_{T\,ICE}/C_{T\,OCE_J}$ at $\theta = 75°$ is smaller than that of $\theta = 90°$ at $J > 0.1$, indicating a much smaller thrust augmentation in the ceiling effect when tilting the propeller forward.

Since normalizing the data with OCE at different J values brings more insight into the effect of the ceiling on the propeller performance, the normalization in the following sections will be performed with the same method. An in-depth discussion of the changes in OCE propeller performance due to propeller incidence angle is included in our previous study [22].

Figure 8 shows the normalized power coefficient, $C_{P\,ICE}/C_{P\,OCE_J}$, for both $\theta = 90$ and 75°. For $\theta = 90°$ and $h/D > -0.21$, $C_{P\,ICE}/C_{P\,OCE_J}$ remain relatively constant with the increment in J and merge with other curves between $0.3 < J < 0.4$. At $-0.07 > h/D > -0.21$, $C_{P\,ICE}/C_{P\,OCE}$ slightly increases with an increment in J before merging with other curves. After merging, $C_{P\,ICE}/C_{P\,OCE}$ increase with further increments of J and then reaches 1.0 at $J = 0.4$, resulting in no ceiling effect power benefit.

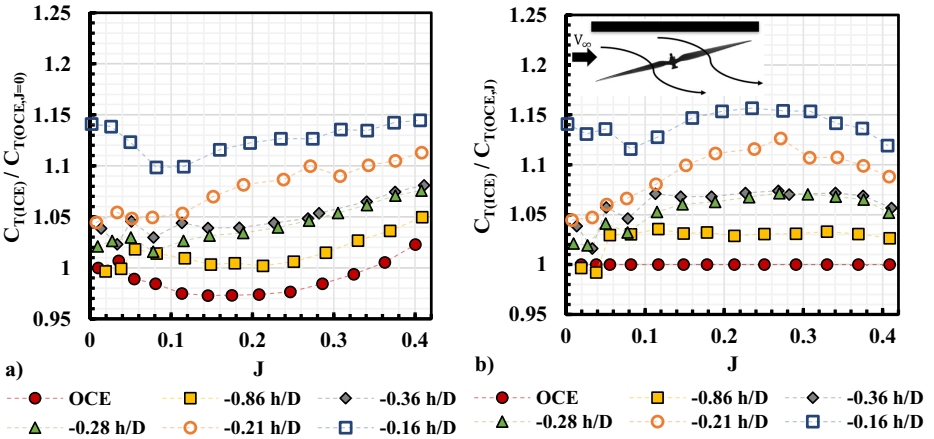


Fig. 7 Normalized thrust coefficient a) $C_{T\,ICE}/C_{T\,OCE_{J=0}}$ and b) $C_{T\ IGE}/C_{T\,OCE_J}$ at $\theta = 75°$ under 100 RPS in ceiling effect for APC 14×6.

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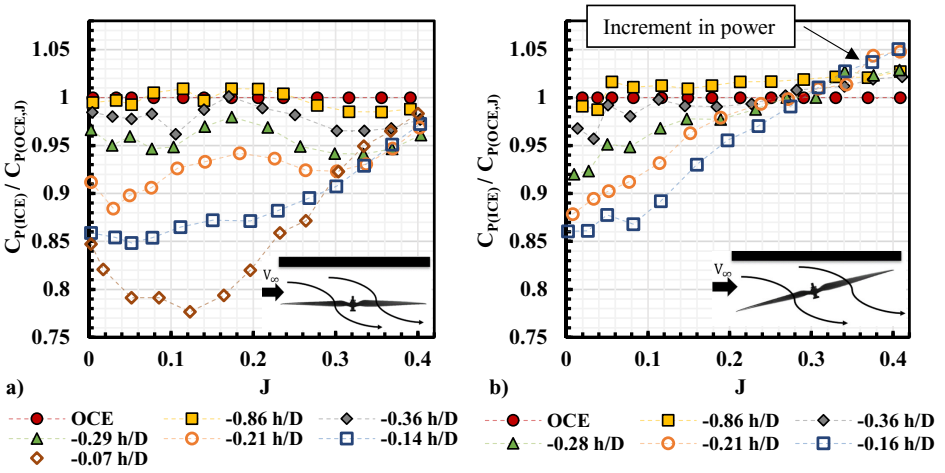


Fig. 8 Normalized power coefficient ($C_{P\,\mathrm{ICE}}/C_{P\,\mathrm{OCE}_J}$) at a) $\theta = 90°$ and b) $\theta = 75°$ under 100 RPS in ceiling effect for APC 14 × 6.

An abnormal decrease of C_p is observed for the extreme ceiling effect cases at $h/D = -0.07$ between $0 < J < 0.12$. This is most likely a result of blade stall recovery, as the distance between the propeller and the ceiling plane is small. At low J, not enough upstream inflow is provided to the propeller, resulting in an increase in the blade effect angle of attack, which leads to a propeller blade stall. With the increment in J, the inflow of the propeller increases, which reduces the rotor effect angle of attack, reducing the local blade drag, and ultimately reducing the power consumption of the propeller.

At $\theta = 75°$, a clear increment in $C_{P\,\mathrm{ICE}}/C_{P\,\mathrm{OCE}_J}$ with the increment of J is observed. All curves merge at $J \sim 0.2$, which is lower than that of $\theta = 90°$. After the curve merges, $C_{P\,\mathrm{ICE}}/C_{P\,\mathrm{OCE}_J}$ continues to increase and reach 1.0 at $J = 0.3$, which is also lower than the $\theta = 90°$ case. At $J > 0.3$, $C_{P\,\mathrm{ICE}}/C_{P\,\mathrm{OCE}_J}$ reaches over 1.0, indicating a higher power consumption at the same rotational speed when compared to the OCE condition.

As the increment in thrust and decrement in power vary simultaneously in the ceiling effect, a common representation of the overall propeller ICE performance in thrust and power is the "normalized power" ($P_{\mathrm{ICE}}/P_{\mathrm{OCE}}$) at constant thrust. To calculate $P_{\mathrm{ICE}}/P_{\mathrm{OCE}_J}$, the propeller ICE and OCE thrust and torque are measured over a range of propeller rotational speeds between 80 and 120 RPS, at the same advance ratio. For every h/D of interest, a second-order interpolation for thrust versus rpm and torque versus rpm across the rotational speed is created. Then, a thrust level of interest is selected, and the rotational speed at the selected thrust level is found using the thrust interpolation at every h/D. At this particular rotational speed, the IGE power is obtained from the torque interpolation. With the freestream velocity known, and propeller rotational speed calculated from the interpolation, the advance ratio is then calculated. The ICE power is normalized by the power of OCE at the same thrust level and advance ratio to produce $P_{\mathrm{ICE}}/P_{(\mathrm{OCE}_J)}$. The detailed calculation method can be found in our previous study [22].

In this section, a thrust level of 15 N is selected to represent the medium thrust level for the APC 14 × 6 propeller and the results are shown in Fig. 9. For $\theta = 90°$, all cases have a small increment in $P_{\mathrm{ICE}}/P_{\mathrm{OCE}}$ at $J < 0.15$, except for the $h/D = -0.07$ case where $P_{\mathrm{ICE}}/P_{\mathrm{OCE}}$ remains constant until $J = 0.3$. The trend is then reversed between $0.15 < J < 0.3$. Since the changes in C_P within the J range are small, the changes in $P_{\mathrm{ICE}}/P_{\mathrm{OCE}}$ are mainly a result of the changes in C_T seen in Fig. 6 at the same J range. At a higher J range, the value of $P_{\mathrm{ICE}}/P_{\mathrm{OCE}}$ increases for all cases due to the loss in ICE power savings seen in Fig. 8.

When tilting the propeller forward at $\theta = 75°$, $P_{\mathrm{ICE}}/P_{\mathrm{OCE}_J}$ remains relatively constant at $J < 0.15$ for all h/D tested. However, the value of $P_{\mathrm{ICE}}/P_{\mathrm{OCE}_J}$ does not reduce between $0.15 < J < 0.3$, except for $h/D = -0.16$ case, where $P_{\mathrm{ICE}}/P_{\mathrm{OCE}}$ continues to increase. The increment in $P_{\mathrm{ICE}}/P_{\mathrm{OCE}_J}$ occurs again at $J > 0.3$ for all h/D cases, as also observed in the $\theta = 90°$ case. Overall, a clear reduction trend of $P_{\mathrm{ICE}}/P_{\mathrm{OCE}_J}$ is observed with the increment in J between $0 < J < 0.4$, indicating a reduction of the ceiling effect benefit at higher forward flight speed. Therefore, the propeller ceiling effect performance in hover cannot be used to estimate the propeller performance in ceiling effect forward flight, especially when the rotor is tilted toward the freestream, where the ICE thrust augmentation reduces while the power consumption increases.

2. *Results* 11 × 7 *Propeller*

Figure 10 shows the $C_{T\,\mathrm{ICE}}/C_{T\,\mathrm{OCE}_J}$ result for a higher γ/D propeller, APC 11 × 7. At $\theta = 90°$, the reduction in $C_{T\,\mathrm{ICE}}/C_{T\,\mathrm{OCE}_J}$

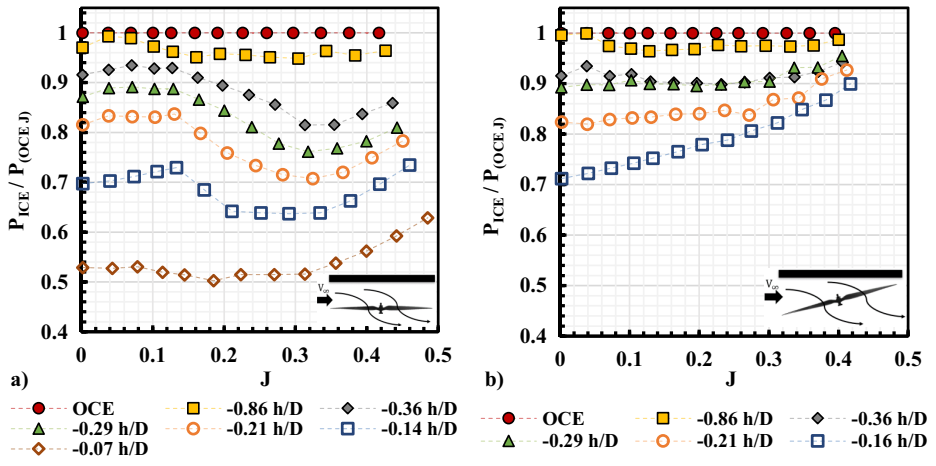


Fig. 9 Normalized power required at constant thrust ($P_{\mathrm{ICE}}/P_{\mathrm{OCE}_J}$) at a) $\theta = 90°$ and b) $\theta = 75°$ in ceiling effect for APC 14 × 6 at 15 N of thrust.

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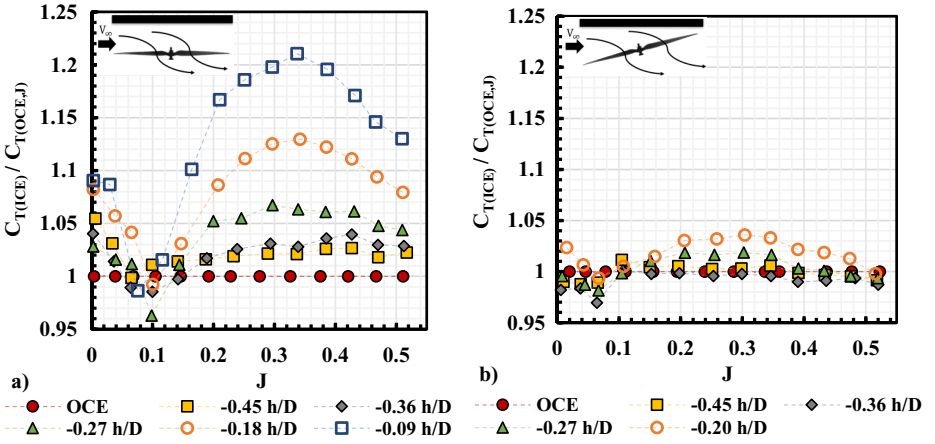


Fig. 10 Normalized thrust coefficient ($C_{T\,\mathrm{ICE}}/C_{T\,\mathrm{OCE}_J}$) at a) $\theta = 90°$ and b) $\theta = 75°$ under 100 RPS in ceiling effect for APC 11×7.

with the increment in J becomes much more severe at $J < 0.1$ when compared to a lower γ/D propeller, the APC 14×6. At $J = 0.1$, $C_{T\,\mathrm{ICE}}/C_{T\,\mathrm{OCE}_J}$ reaches below 1.0, indicating that the ICE thrust augmentation observed in hovering vanishes. Further increments in J result in an increment in $C_{T\,\mathrm{ICE}}/C_{T\,\mathrm{OCE}_J}$ between $0.1 < J < 0.3$, indicating a recovery of the ICE thrust augmentation in forward flight. This increment once again is much more severe when compared to the APC 14×6 case. At $J > 0.3$, $C_{T\,\mathrm{ICE}}/C_{T\,\mathrm{OCE}_J}$ decreases with the increment in J again, which is similar to the trend seen for the APC 14×6 propeller in Fig. 7.

When tilting the propeller forward at $\theta = 75°$, ICE thrust benefit in hover ($J = 0$) reduces significantly for the APC 11×7 propeller when compared to the $\theta = 90°$ case, which is different from the APC 14×6 propeller, where the ICE thrust benefit in hover remains the same between $75° < \theta < 90°$. Similar to the $\theta = 90°$, $C_{T\,\mathrm{ICE}}/C_{T\,\mathrm{OCE}_J}$ reduces with the increment of J and reaches ~1.0 at $J = 0.1$. $C_{T\,\mathrm{ICE}}/C_{T\,\mathrm{OCE}_J}$ then reduces between $0.1 < J < 0.3$ and increases at $J > 0.3$ with the increment in J, though the magnitude is much smaller than the $\theta = 90°$ case, indicating a much smaller ICE benefit on thrust generation.

$C_{P\,\mathrm{ICE}}/C_{P\,\mathrm{OCE}_J}$ for the APC 11×7 propeller at both θ cases is shown in Fig. 11. At $J < 0.1$, $C_{P\,\mathrm{ICE}}/C_{P\,\mathrm{OCE}_J}$ remains relatively constant, except for the extreme ceiling effect case at $h/D = -0.09$, where a decrement in $C_{P\,\mathrm{ICE}}/C_{P\,\mathrm{OCE}_J}$ is observed, similar to that of the APC 14×6 propeller in Fig. 8. At $J > 0.1$, $C_{P\,\mathrm{ICE}}/C_{P\,\mathrm{OCE}_J}$ increases with the increment in J, reaching1.0 at $J \sim 0.2$, and increases beyond 1.0 at $J > 0.2$, indicating a higher ICE power consumption. The normalized C_P never reaches 1.0 with increase in J. For $\theta = 75°$, the results are very similar to the APC 14×6 propeller where $C_{P\,\mathrm{ICE}}/C_{P\,\mathrm{OCE}_J}$ increases with the increment of J and reaches 1.0 at $J \sim 0.25$, slightly lower than the APC 14×6 propeller. $C_{P\,\mathrm{ICE}}/C_{P\,\mathrm{OCE}_J}$ then remains ~1.0 at higher J, indicating a minimal ICE effect on the propeller power consumption for a higher γ/D propeller.

Similar to the 14×6 propeller, $P_{\mathrm{ICE}}/P_{\mathrm{OCE}_J}$ for the 11×7 propeller is calculated to better represent the ICE performance. The results are shown in Fig. 12. A clear increment in $P_{\mathrm{ICE}}/P_{\mathrm{OCE}_J}$ is observed at $J < 0.1$ with the increment in J at $\theta = 90°$. This is mainly due to the reduction in the propeller thrust generation seen in Fig. 10. $P_{\mathrm{ICE}}/P_{\mathrm{OCE}_J}$ approaches almost 1.0 at $J = 0.1$, indicating a complete loss of the ICE performance benefit. At $0.1 < J < 0.3$, $P_{\mathrm{ICE}}/P_{\mathrm{OCE}}$ reduces and reaches a similar performance as hovering ($J = 0.0$) at $J = 0.3$. Similar to the lower γ/D propeller, $P_{\mathrm{ICE}}/P_{\mathrm{OCE}_J}$ increases with further increment of J, indicating a reduction of the ICE performance benefit. At $\theta = 75°$, the trend of $P_{\mathrm{ICE}}/P_{\mathrm{OCE}_J}$ resembles the case of the lower γ/D propeller, where $P_{\mathrm{ICE}}/P_{\mathrm{OCE}_J}$ increases with the increment in J. It is worth noticing that at $J > 0.4$, $P_{\mathrm{ICE}}/P_{\mathrm{OCE}_J}$ reaches above 1.0, indicating higher ICE power consumption when generating a constant thrust compared to the OCE condition for the higher γ/D propeller. Therefore, when selecting the propeller for the constant in-ceiling effect vehicle, a higher γ/D propeller should be avoided.

3. *Overall ICE Performance in Forward Flight*

To reflect the real flight conditions, the power required at constant thrust for both 14×6 and 11×7 propellers is normalized by the OCE condition at hover, resulting in $P_{\mathrm{ICE}}/P_{\mathrm{OCE}_{J=0}}$, which accounts for the effect of freestream velocity on the propeller performance. The selected h/D result for both APC 14×6 and APC 11×7 propellers at $\theta = 90$ and 75° is shown in Fig. 13. As mentioned previously, the

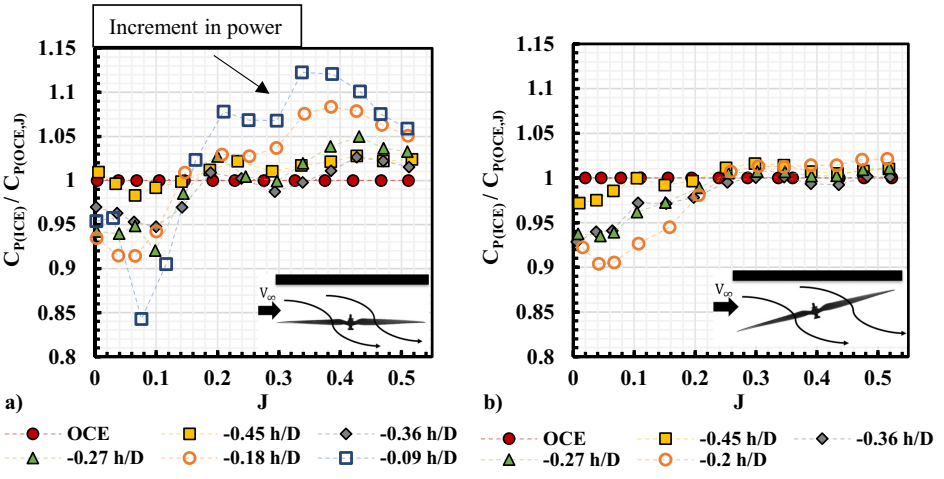


Fig. 11 Normalized power coefficient ($C_{P\,\mathrm{ICE}}/C_{P\,\mathrm{OCE}_J}$) at a) $\theta = 90°$ and b) $\theta = 75°$ under 100 RPS in ceiling effect for APC 11×7.

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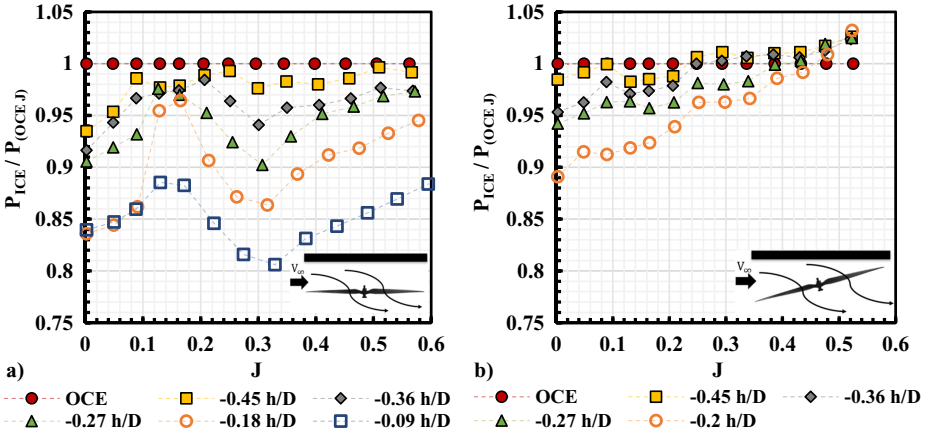


Fig. 12 Normalized power required at constant thrust (P_{ICE}/P_{OCE_J}) at a) $\theta = 90°$ and b) $\theta = 75°$ in ceiling effect for APC 11 × 7 at 8 N of thrust.

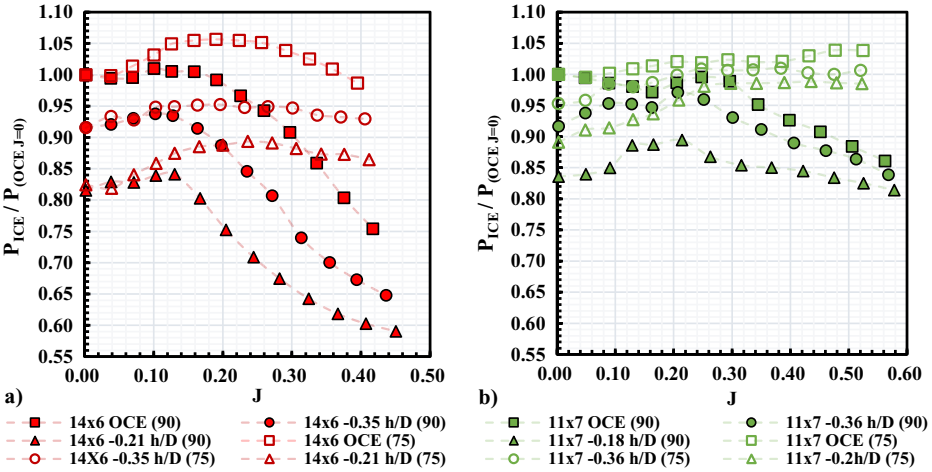


Fig. 13 OCE hover normalized power required at constant thrust for a) APC 14 × 6 (15N) and b) APC 11 × 7 (8N) propeller at $\theta = 90°$ and $\theta = 75°$.

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ICE benefit in thrust augmentation and power saving due to the proximity to the ceiling plate reduces significantly when the propeller is tilted forward to achieve the forward flight condition; this is also shown in Fig. 13. This suggests that the ICE performance of the propeller under hover conditions cannot be used to represent the propeller's performance in forward flight, especially when the propeller is tilted, as it would lead to an overestimation of the propeller's performance.

For different propeller geometries, the overall ceiling effect benefit for the propeller with higher γ/D (APC 11 × 7) is smaller compared to the lower γ/D propeller (APC 14 × 6). This is because $P_{ICE}/P_{OCE_{J=0}}$ for the higher γ/D propeller is always greater than that of the propeller with lower γ/D under the same h/D. Moreover, the reduction in the ICE benefit is more pronounced for a higher γ/D propeller in forward flight, especially at $\theta = 75°$, where $P_{IGE}/P_{OGE_{J=0}}$ reaches ∼1.0 at $J \geq 0.2$, while $P_{IGE}/P_{OGE_{J=0}}$ remains at ∼0.9 for the lower γ/D propeller at $h/D > -0.35$. Again, this underscores the importance of the propeller selection when designing a vehicle to operate under constant in-ceiling-effect in forward flight.

Despite the difference in the magnitude of $P_{ICE}/P_{OCE_{J=0}}$ seen in Fig. 13, the overall trend in P_{ICE}/P_{OCE} versus J is very similar. This similarity is also evident when normalizing the power required at constant thrust versus J at different h/D by the corresponding powered required at $J = 0$, resulting $P_{ICE}/P_{ICE_{(J=0)}}$. The experimental results for the APC 14 × 6 propeller in Fig. 9 are replotted in terms of $P_{ICE}/P_{ICE(J=0)}$ in Fig. 14. For both θ cases, $P_{ICE}/P_{ICE_{(J=0)}}$ aligns with the OCE curve, except for the lowest h/D case, where the propeller disk is extremely close to the ceiling plate, resulting in a higher $P_{ICE}/P_{ICE_{(J=0)}}$. This suggests that the propeller ICE performance can be predicted using only the propeller ICE performance at $J = 0$, which can be estimated by the model developed in our previous study [18] on two-blade propellers ($\gamma/D < 0.7$) with an error of less than 6%.

Additionally, the propeller OCE performance at the corresponding θ in forward flight can be estimated based on the propeller performance at $\theta = 0°$ [27,28], as well as a specialized panel method code with a small error [29]. Therefore, when predicting the propeller ICE performance in the ceiling effect, consideration should be given not only to the propeller ICE performance in hover but also to the propeller performance in OCE forward flight. In general, a fixed pitch propeller with lower γ/D would have a larger benefit in ceiling effect. Meanwhile, a lower γ/D propeller also has better performance in edgewise forward flight [28,30]. Hence, when designing a vehicle operating in the ceiling effect, a lower γ/D propeller is preferred.

B. Ceiling Effect Phase-Lock PIV

A phase-locked PIV experiment was conducted to examine the flowfield around the APC 14 × 6 propeller in the ceiling effect. The images were captured at the same propeller phase, where the blade rotates about 80 deg away from the laser sheet. The contours presented in this section are averaged over several instantaneous images acquired when the propeller blade was at the same location (phase) in every six rotations. The vortex centers of the propeller tip vortex were tracked for each image based on the maximum Q criterion. If the vortex center's location is outside the 95% confidence interval on either axis, the image is excluded from the analysis. In cases where the propeller tip vortex is not identifiable and is out of the FOV, the propeller inflow uniformity is used to remove the outlier images. During this process, an average of 4.7% of the images were eliminated. The measured vorticity in the PIV image is normalized by the

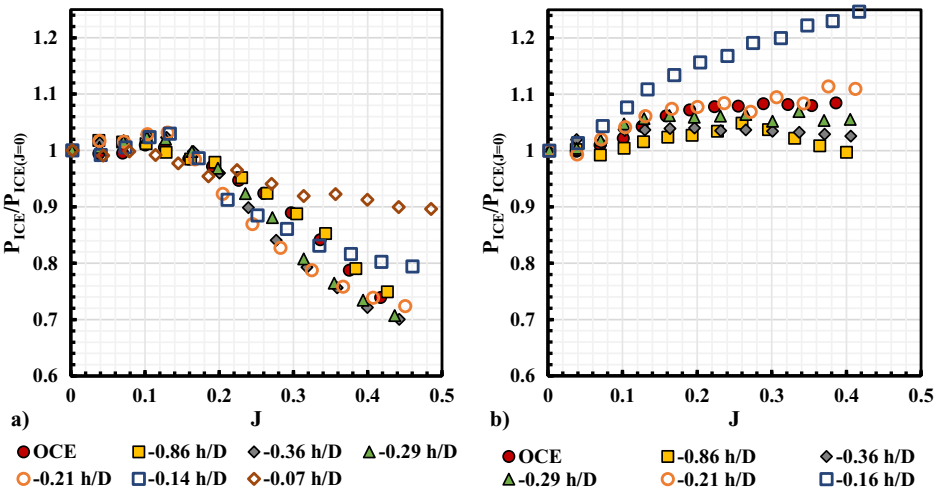


Fig. 14 Normalized power required at constant thrust ($P_{ICE}/P_{ICE_{J=0}}$) at a) $\theta = 90°$ and b) $\theta = 75°$ in ceiling effect for APC 14 × 6 at 15 N of thrust.

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propeller diameter and the propeller blade speed at a 75% radius location, as shown in Eq. (3).

$$\omega_{\text{norm}} = \frac{\omega * D}{n * \pi D * 0.75} \tag{3}$$

Figure 15 shows ω_{norm}, the velocity vectors, and the streamline from the PIV results on the leading side of the propeller disk (FOV A in Fig. 3), as well as the trailing side (combination of FOV B and C in Fig. 3) for $h/D = -0.16$ case, providing an overview of the flowfield in the ICE condition. The freestream flows from left to right, while the ceiling plate is located at the top of the FOV. The propeller size and the distance from the ceiling are shown to scale. The flowfield near the leading edge (180° blade azimuthal angle) and trailing edges (0° blade azimuthal angle) of the propeller disk is symmetric at $J = 0$ due to the absence of freestream velocity. The magnitudes of the tip vortex vorticity on the leading and trailing edge sides are opposite, indicating counter-rotational directions as expected. As J increases to 0.1, the flowfield remains nearly symmetric, with the propeller's wake momentum still predominating. When J exceeds 0.2, the flowfield becomes asymmetric, exhibiting complex flow patterns, including a region of reverse flow and a nodal point on the trailing side of the propeller disk. At the highest J tested, the freestream momentum dominates, and the flowfield on both the leading and trailing sides of the propeller disk resembles the OCE condition, which will be discussed in the next section.

In the subsequent sections, the discussions will concentrate on comparing the flowfield between different ICE and OCE cases, examining the leading and trailing sides of the propeller disk separately.

1. *Effect of Ceiling on Propeller Inflow at Disk Leading Side*

Figure 16 depicts the PIV results on the leading side of the propeller disk for $\theta = 90°$ at $h/d = -0.16$, $h/d = -0.29$, and OCE condition. The origin of the coordinate system for all PIV images is the hub of the propeller. The contour plot illustrates the propeller inflow angle φ along with the phase-averaged velocity vector of the flowfield. An inflow angle of $\varphi = 0°$ represents inflow from the axial direction of the propeller, whereas $\varphi = 90°$ represents inflow from the radial direction, a scenario similar to the edgewise flight condition. As the propeller inflow is not uniform along the radius of the propeller, the average propeller inflow angle $\overline{\varphi}$ is used to represent the overall inflow conditions of the propeller. To calculate $\overline{\varphi}$, the inflow angle at a distance equivalent to one propeller chord (chord length at 75% r/R) above the propeller disk, between $0.5 < r/R < 0.8$, is averaged based on the PIV results. The φ values for the selected cases are shown in Fig. 17.

At $J = 0.0$, where the propeller is in a hovering condition, a higher φ is observed for the ICE condition compared to the OCE condition. For $h/d = -0.16$, $\overline{\varphi}$ increased to 69.8° when compared to the OCE where $\overline{\varphi} = 27.4°$. This increase is attributed to the presence of the ceiling plate, which prevents the propeller inflow from approaching the propeller disk along its axial direction, necessitating that the inflow comes from the radial direction of the propeller disk, thus increasing $\overline{\varphi}$. Consequently, the propeller blade operates at a higher effective angle of attack, akin to the edgewise forward flight condition, which is observed to result in an increase in propeller thrust generation, due to a higher effective angle of attack experienced by the propeller blade [28,30]. This is consistent with the C_T measurement is shown in Fig. 6, which indicates an increase in C_T during hovering at low h/D values.

Increasing h/D results in a modest increase in $\overline{\varphi}$ to 51.2°, which signifies a diminishing impact of the ceiling plane on propeller performance. As the distance between the propeller disk and the ceiling plane widens, the inflow velocity in the radial direction decreases, assuming that the induced velocity at the propeller plane remains relatively constant at the same rotational speed. Figure 16 corroborates this, as it shows a smaller velocity vector at $J = 0.0$ for $h/D = -0.29$ at $x = 0.1$ compared to $h/D = -0.16$. The lower $\overline{\varphi}$ results in a reduced C_T generated by the propeller, which is also evident in Fig. 6.

When J increases to 0.1 under the OCE condition, the radial inflow to the propeller increases due to the presence of the freestream, resulting in a higher $\overline{\varphi}$. However, as the propeller wake momentum still dominates at $J = 0.1$, the influence of the freestream is relatively minor, which holds true for the ICE condition where the variation in C_T is relatively small between $0 < J < 0.1$. As J increases, $\overline{\varphi}$ further increases for all cases. At $J = 0.2$, the $\overline{\varphi}$ values for $h/D = -0.16$, $h/D = -0.29$ and OCE condition are 82.2, 73.4, and 63.1°, respectively. Once again, a lower h/D implies a greater effect on the ceiling plane on the propeller inflow angle, and consequently, a greater ICE benefit.

As the freestream velocity continues to increase, $\overline{\varphi}$ increases as well. However, the disparity in $\overline{\varphi}$ decreases, especially at $J = 0.4$, where $\overline{\varphi}$ for the $h/D = -0.16$, $h/D = -0.29$ and OCE condition are 88.6, 85.0, and 80.0°, respectively. As the ICE $\overline{\varphi}$ approaches the OCE value condition at a higher J, and as the flowfield becomes more similar to the OCE condition, the overall ICE benefit declines. This is also reflected in the power required at constant thrust performance, as shown in Fig. 13, where a decrease in P_{ICE}/P_{OCE} is observed for $J > 0.3$.

With a lower incidence angle of $\theta = 75°$ (Fig. 18), the overall trend in inflow is very similar to the $\theta = 90°$ cases. At $J = 0.0$, the ceiling plane continues to exert a significant influence on the propeller inflow, with $\overline{\varphi} = 75$ and 56°, compared to a $\overline{\varphi}$ of 51° for the OCE condition. Similarly, an increment in J result in an increase in $\overline{\varphi}$. At the highest J, the $\overline{\varphi}$ values for $h/D = -0.16$, $h/D = -0.29$, and OCE condition are 94.5, 90.5, and 88°, respectively. It is worth noting that $\overline{\varphi}$ for the $h/D = -0.29$ case is nearly identical to the OCE condition. Referencing Fig. 7, the thrust augmentation for the

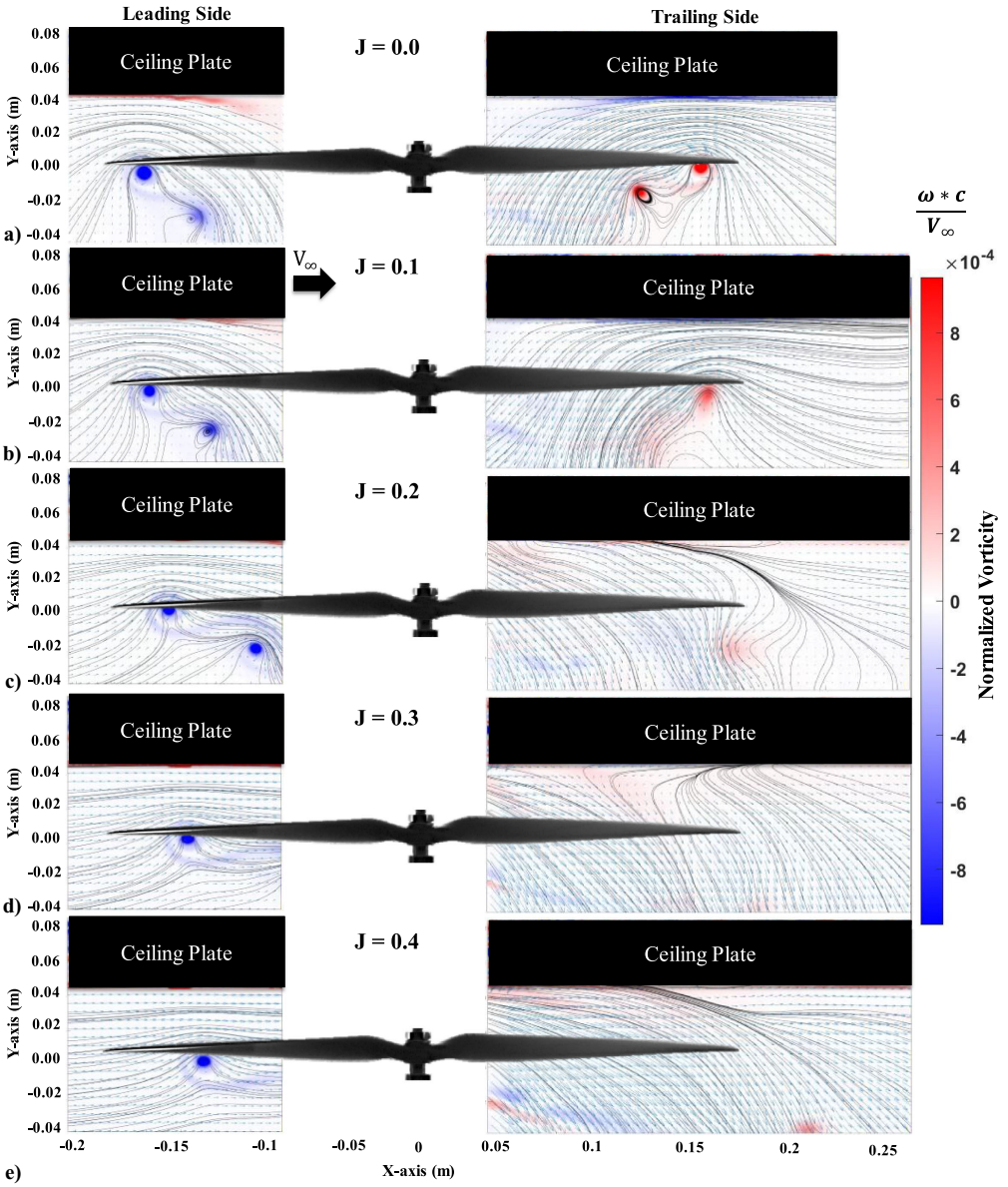


Fig. 15 PIV results for propeller leading (left) and trailing (right) side at $h/D = -0.16$ under a) $J = 0.0$, b) $J = 0.1$, c) $J = 0.2$, d) $J = 0.3$, and e) $J = 0.4$.

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$h/D = -0.29$ case at $J > 0.3$ is less pronounced than at lower J conditions. In comparison with results in Figs. 13 and 16, the impact of the ceiling plane on $\overline{\varphi}$ also reduces with a decrease in propeller incidence angle. As the distance between the propeller tip on the disk's leading side and the ceiling plane increases, the propeller-induced inflow in the radial direction decreases, leading to a less significant change in the propeller inflow angle. This is analogous to increasing the distance between the propeller and the ceiling plane, as seen in Fig. 16. In this case, the propeller experiences a diminished ICE benefit, a trend that is also observed in Fig. 9.

2. *Effect of the Ceiling on Propeller Inflow at Disk Trailing Side*

A more complex flowfield is evident at the trailing side of the propeller disk. To elucidate the flowfield, the normalized vorticity contour is shown alongside the streamline and the instantaneous velocity vectors of the flowfield, similar to Fig. 15. It is worth noting that the results are a combination of FOV 2 and FOV 3, as a single camera's FOV is incapable of capturing the entire flowfield of interest. The results from both $\theta = 90°$ cases are shown in Fig. 19. At $J = 0.1$, where the propeller's momentum dominates, the flowfield at the trailing side of the propeller disk moves from the tip toward the hub for $h/D = -0.16$, even though the freestream velocity persists. This results in a reverse flow region, and the propeller performance closely resembles the hovering condition, as seen in Fig. 15a. As the distance to the ceiling plate increases, a stagnation point (indicated by a green dot) emerges on the ceiling plane, creating a divergence in flow direction on either side. This results in a decrease in the overall propeller inflow angle at the trailing side of the propeller disk, leading to a slight reduction in the propeller thrust generation, as seen in Fig. 6. This effect mirrors the phenomenon discussed in Fig. 16, where a decrease in the propeller inflow angle leads to a lower C_T.

Upon increasing the advance ratio to $J > 0.1$, the reverse flow region vanishes along with the stagnation point in both h/D cases due to the increase in freestream momentum. A nodal point (highlighted by a yellow dot) is observed, at which the flow velocity is nearly zero. Figure 20 shows the vorticity contour and the phase-averaged velocity vectors for the $\theta = 90°$, $h/D = -0.29$, and $J = 0.3$ case to highlight the presence of the nodal point. A contour line representing a low velocity (0.5 m/s) is drawn first (marked in green); within the region bounded by this contour line and the ceiling plate, the flow velocity is almost zero. Since the direction of the flow diverges on either side of this point, it is designated as the nodal point. Upstream of the nodal point, the flowfield does not resemble the OCE

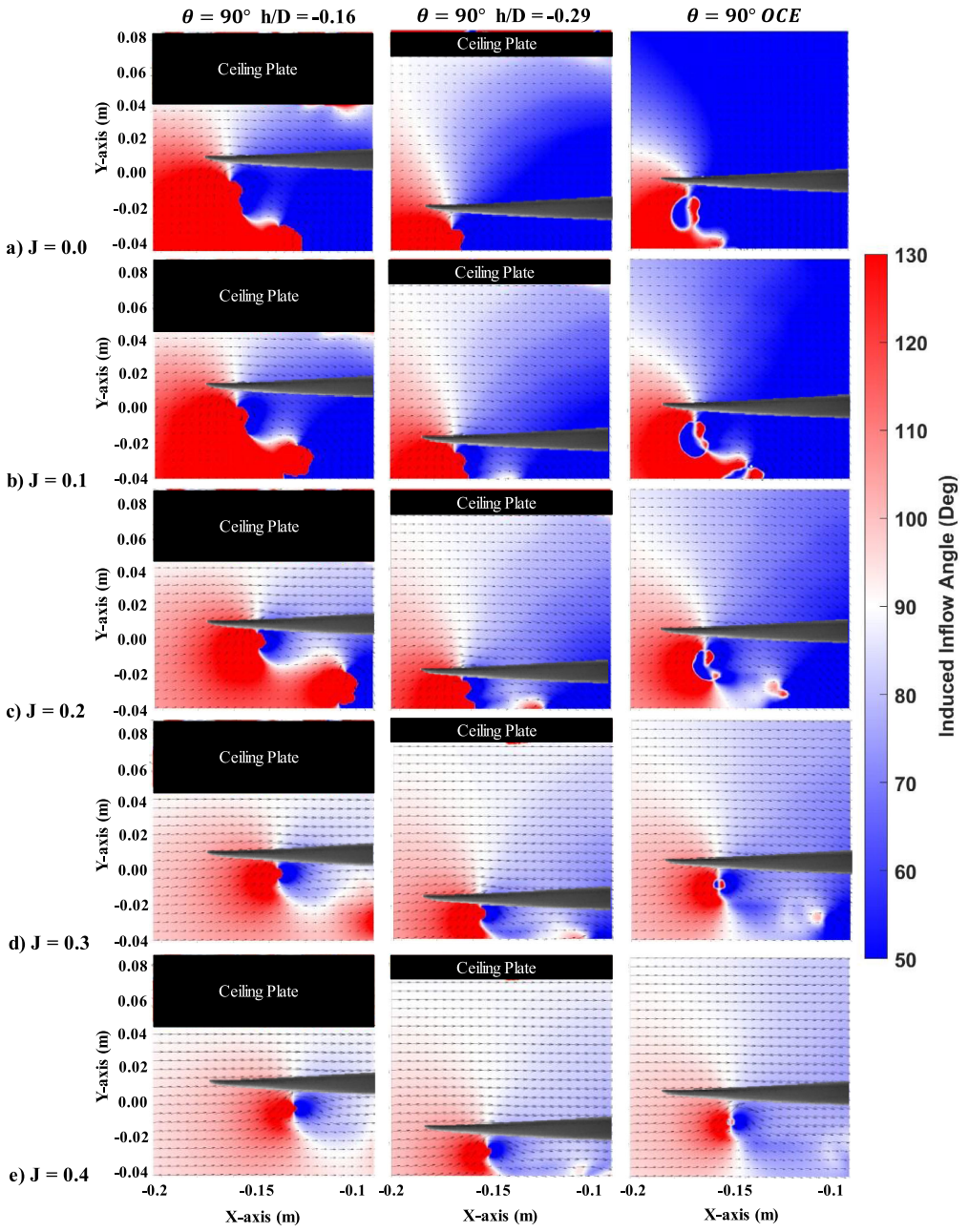


Fig. 16 PIV results at $\theta = 90°$ near propeller disk leading side and $h/D = -0.16$ (left), $h/D = -0.29$ (middle), and OCE condition (right) under a) $J = 0.0$, b) $J = 0.1$, c) $J = 0.2$, d) $J = 0.3$, and e) $J = 0.4$.

condition. Further upstream, a boundary is reached where the upstream propeller inflow becomes uniform and the vorticity magnitude approaches zero. The propeller inflow streamline is relatively uniform ahead of this boundary, similar to the OCE condition. Extending this boundary upstream reveals another nodal point at the junction where the boundary meets the ceiling plane. However, this point is not captured within the camera's FOV. Between these two nodal points, the streamline exhibits a complex pattern due to the out-of-plane inflow (marked by a yellow dashed line). This observation aligns with insights gained from smoke and surface oil flow visualization, which will be discussed in detail in the subsequent section.

At $J = 0.2$ and $h/D = -0.16$, where the distance between the propeller disk and ceiling plane is smaller than the OCE condition, the inflow from the upstream freestream is inadequate for the propeller's operating condition. Consequently, a low-pressure zone forms between the propeller and the ceiling plane, driving additional out-of-plane flow from the sides of the propeller disk toward its center, between the two aforementioned nodal points. The region of the propeller blade affected by the out-of-plane inflow (indicated by a red dashed box in Fig. 19) experiences a higher inflow angle, thereby enhancing its ICE performance. By increasing the distance between the propeller and the ceiling plate, a larger segment of the propeller blade is subjected to uniform flow from upstream. As a result, the region where the propeller blade is influenced by the ceiling plane and the out-of-plane flow reduces, leading to a reduced ICE propeller thrust augmentation, which is consistent with the C_T measurement in Fig. 6.

Further increasing J to 0.3 results in the nodal point moving downstream for both cases. This is strikingly similar to the phenomena observed in the ground effect flowfield in our previous study [22], where a stagnation point on the ground plate moves downstream as J increases due to the increase in freestream momentum. As the nodal point shifts downstream of the propeller disk and the low-pressure zone between the propeller disk and ceiling plane drives out-of-plane flow toward the propeller disk, a reverse flow region is observed between the propeller disk and the ceiling plate at $h/D = -0.16$. This reverse flow region induces an even higher inflow angle for the propeller blades within this area, further enhancing the propeller

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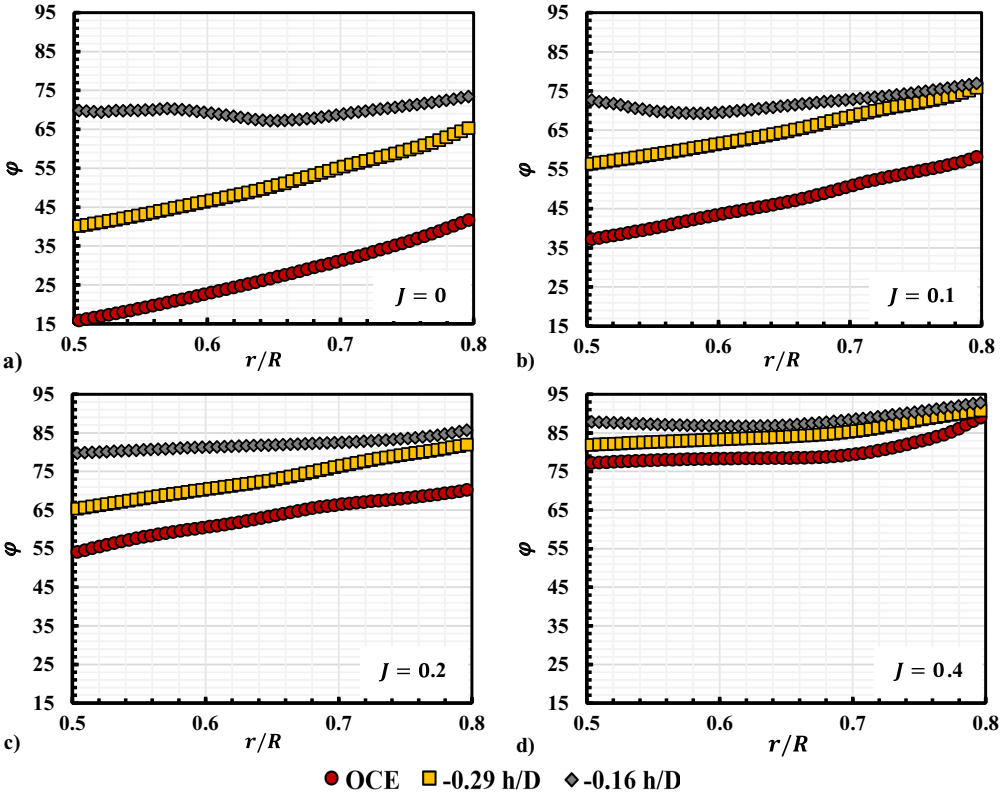


Fig. 17 Distribution of φ between $0.5 < r/R < 0.8$ at $\theta = 90°$ under a) $J = 0.0$, b) $J = 0.1$, c) $J = 0.2$, and d) $J = 0.4$.

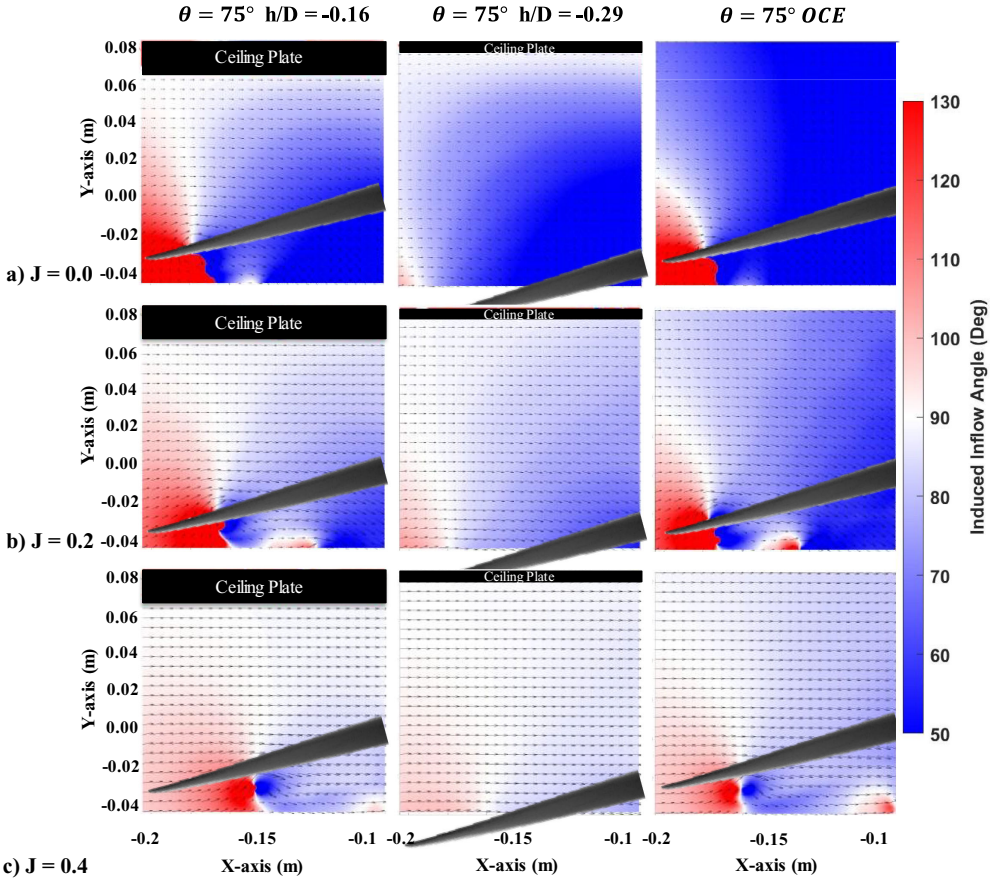


Fig. 18 PIV results at $\theta = 75°$ near propeller disk leading side and $h/D = -0.16$ (left), $h/D = -0.29$ (middle), and OCE condition (right) under a) $J = 0.0$, b) $J = 0.2$, and c) $J = 0.4$.

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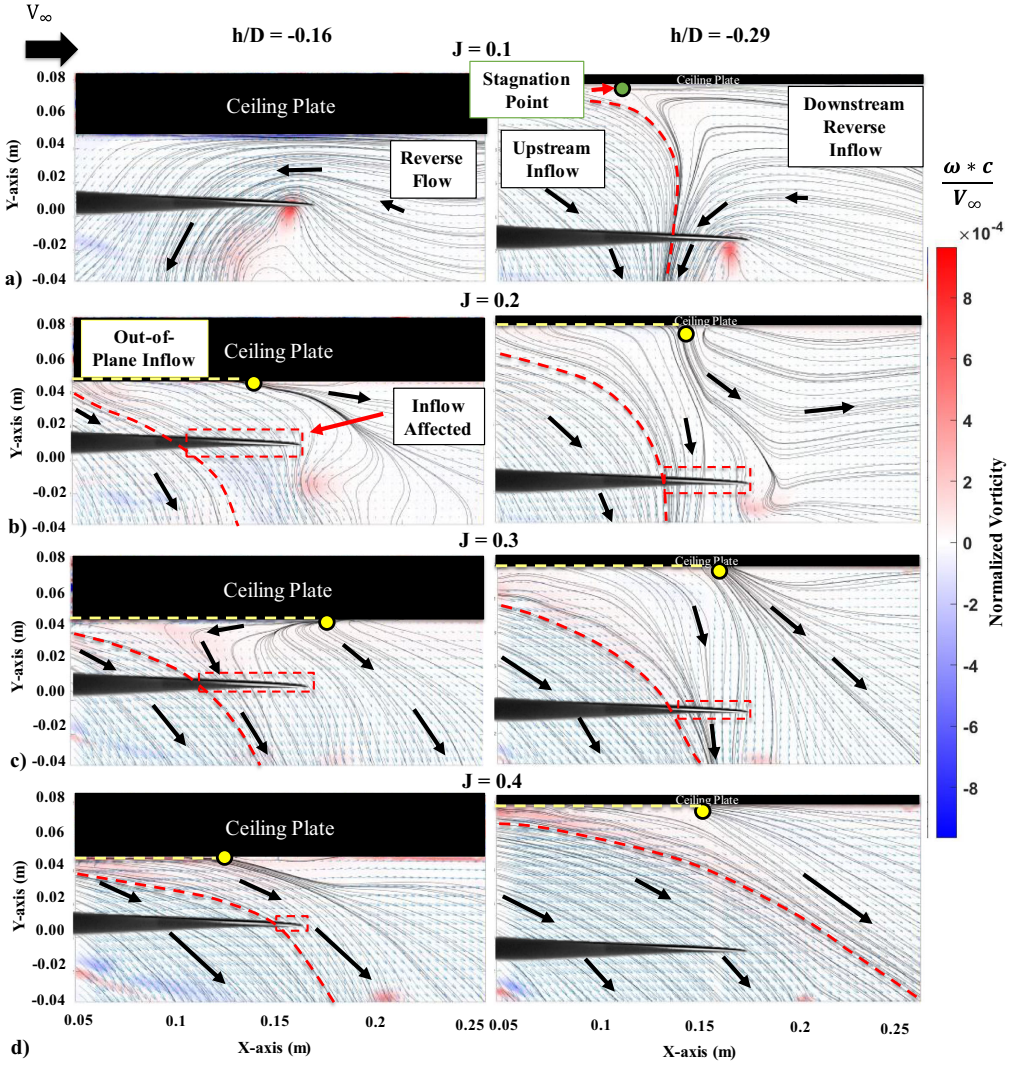


Fig. 19 PIV results at $\theta = 90°$ near propeller disk trailing side and $h/D = -0.16$ (left), $h/D = -0.29$ (right) under a) $J = 0.1$, b) $J = 0.2$, c) $J = 0.3$, and d) $J = 0.4$.

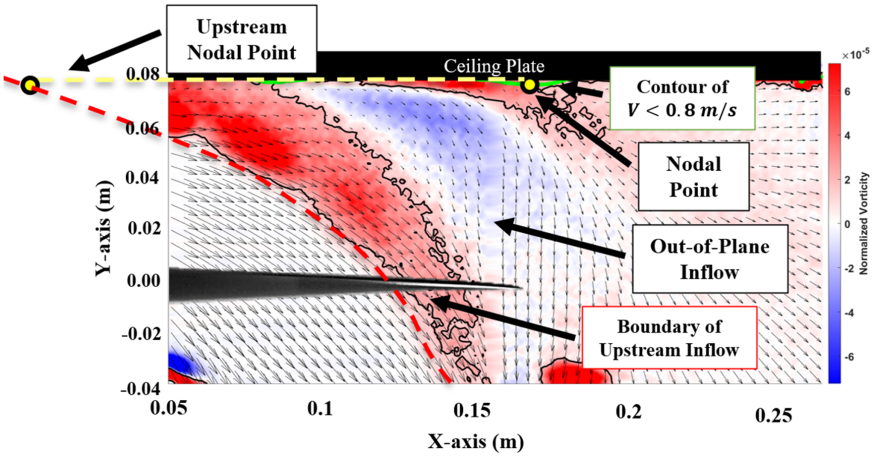


Fig. 20 Vorticity contour and velocity vectors for $\theta = 90°$, $h/D = -0.29$, and $J = 0.3$ showing the nodal point and upstream propeller inflow boundary.

thrust generation, as evidenced in Fig. 6, where $C_{T\,\mathrm{ICE}}/C_{T\,\mathrm{OCE}}$ reaches its peak at $J = 0.3$. It is important to note that the out-of-plane flow is not accurately represented by the PIV result due to the limitation of the 2-D PIV experiment.

Similar to the $J = 0.2$ case, increasing the distance between the propeller and the ceiling plate results in a smaller portion of the propeller blade being affected by the out-of-plane flow, and consequently a diminished ICE performance benefit is observed. At the highest J tested, the nodal point disappears for the $h/D = -0.16$ case, and almost no out-of-plane flow affects the propeller blades. The propeller inflow closely resembles the OCE condition, which is also observed in the $h/D = -0.29$ case. Hence, the thrust augmentation

attributable to the out-of-plane inflow ceases, and the propeller experiences a reduced ICE performance benefit, which is consistent with the results in Fig. 9.

Figure 21 shows the result for the $\theta = 75°$ case, illustrating the impact of the propeller's incidence angle on its ICE performance. At $J = 0.1$ and $h/D = -0.16$, a stagnation point on the ceiling plate is observed closer to the propeller tip for both h/D cases, due to the reverse flow region driven by the propeller. Additionally, a nodal point is detected downstream of the propeller disk in both cases, signifying the presence of the out-of-plane propeller inflow. The propeller guides the downstream out-of-plane flow upstream, leading to the formation of a reverse flow region, analogous to the $\theta = 90°$ case in Fig. 19.

When increasing the advance ratio to $J = 0.2$, the stagnation point persists for the $h/D = -0.16$ case due to the propeller tip's proximity to the ceiling plate. The tip vortex induces a small reverse flow region, which converges the out-of-plane flow from upstream at the stagnation point. The nodal point also shifts onto the ceiling plate for both cases. In alignment with the pattern seen in the $\theta = 90°$ cases, as J or the distance to the ceiling plate increases, the portion of the blade affected by the out-of-plane inflow diminishes due to the increase in the propeller upstream inflow. At $J \geq 0.3$, the influence of the out-of-plane flow on the propeller blade reduces further. Coupled with the trend seen on the leading side of the propeller disk, where the propeller inflow angle approximates the OCE condition, the overall ICE advantage for the propeller performance becomes negligible for $\theta = 75°$ at higher forward flight speeds, corroborating the results from the force-based experiment in Figs. 9 and 13.

3. *Ceiling Effect Smoke and Surface Oil Flow Visualization*

To better understand the ICE propeller flowfield, smoke flow visualization and surface oil flow visualization are conducted on the APC 14×6 propeller at $h/D = -0.16$ and various J. Figure 22 demonstrates the relative perspective of the PIV experiment and both flow visualizations. As mentioned previously, the PIV image shows the propeller flowfield on the YZ-plane. For the smoke visualization, the smoke wand is placed upstream of the propeller disk, and the laser plane is placed on the XZ-plane, which is 90 deg offset from the PIV laser plane, as shown in Fig. 4. The camera is placed downstream of the propeller disk. In this case, the camera was able to capture the flowfield on the YZ-plane that better represents the out-of-plane inflow of the propeller discussed in the PIV results. The smoke flow visualization confirms the existence of the out-of-plane inflow in the ceiling effect. However, it does not provide sufficient information on the out-of-plane inflow. Thus, the discussion will focus on the oil flow visualization results.

For the surface oil flow visualization, the oil mixture is applied to the ceiling plate on the XY plane, the camera is mounted on top of the propeller disk to capture the flowfield on the ceiling plate. The results for the surface oil flow visualization for the APC 14×6 propeller at

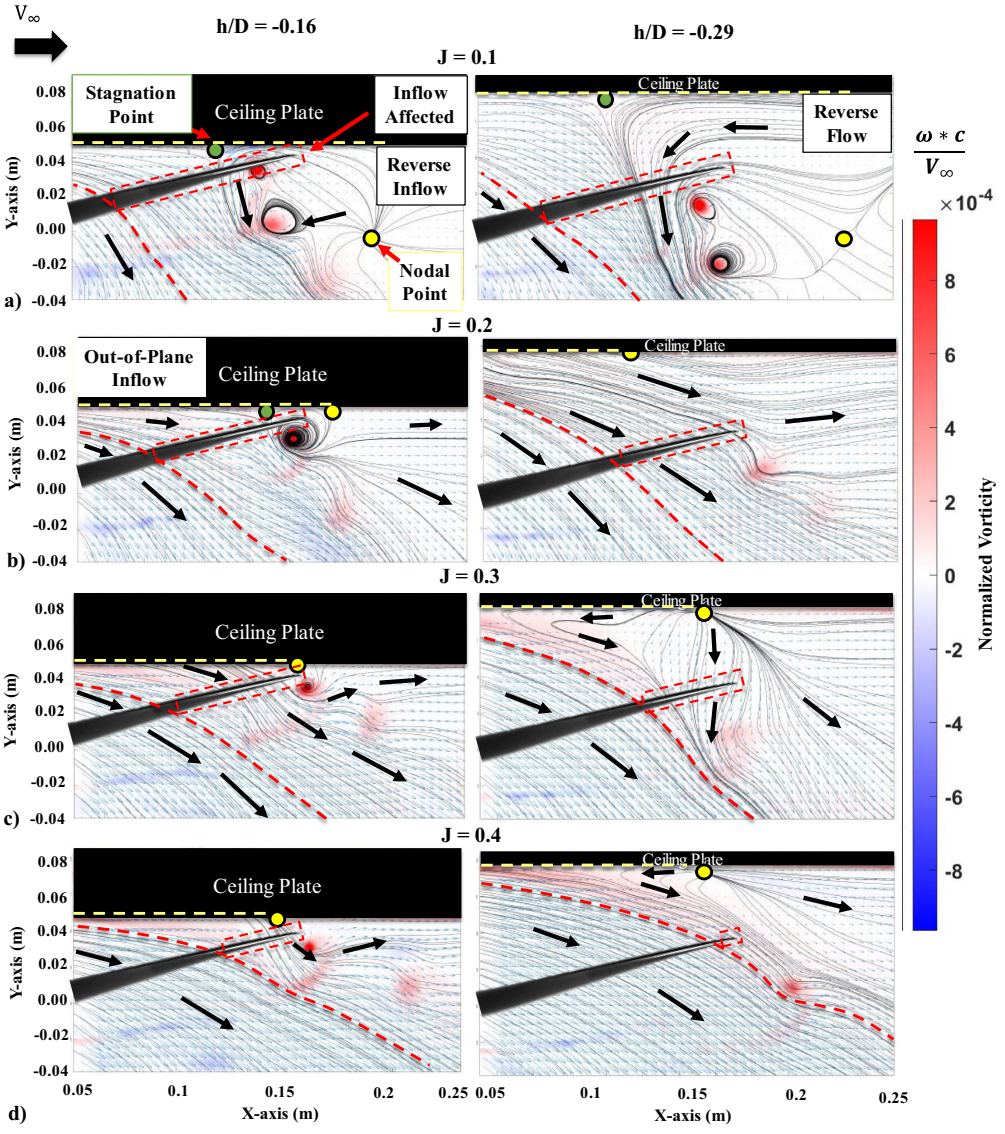


Fig. 21 PIV results at $\theta = 75°$ near propeller disk trailing side and $h/D = -0.16$ (left), $h/D = -0.29$ (right) under a) $J = 0.1$, b) $J = 0.2$, c) $J = 0.3$, and d) $J = 0.4$.

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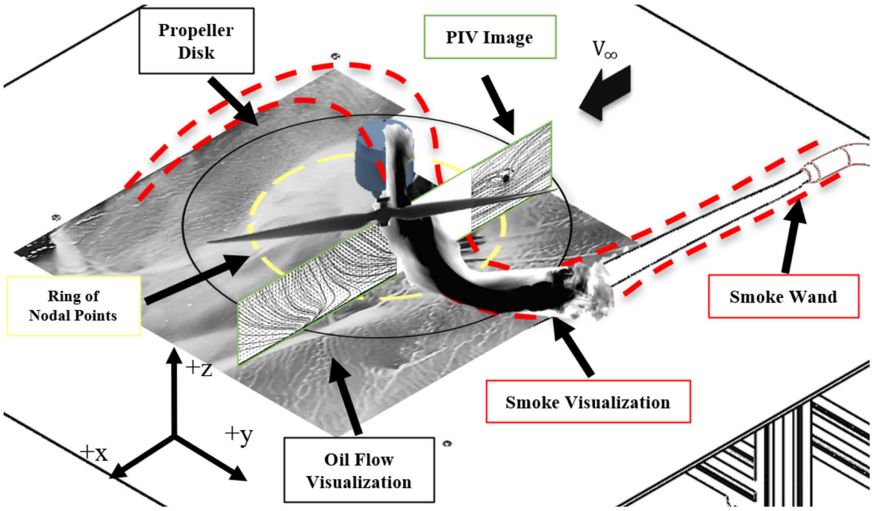


Fig. 22 Isometric view showing the relative perspective of the PIV experiment, smoke visualization, and surface oil flow visualization to the propeller in the ceiling effect.

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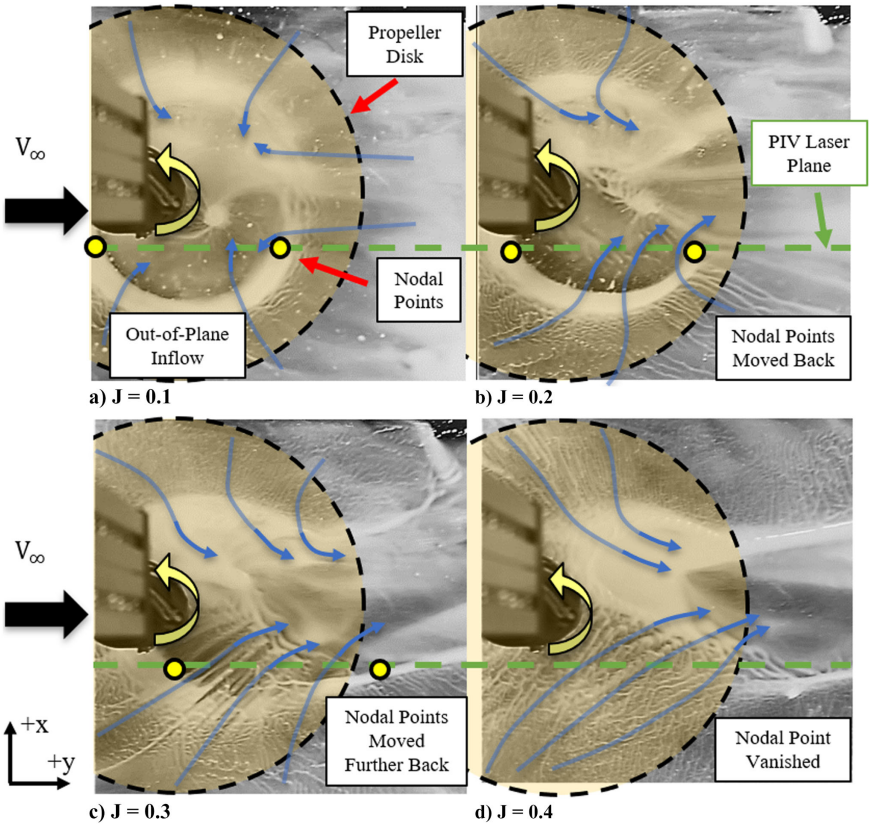


Fig. 23 Surface oil flow visualization for the APC 14 × 6 propeller at $h/D = -0.16$, $\theta = 90°$ under a) $J = 0.1$, b) $J = 0.2$, c) $J = 0.3$, and d) $J = 0.4$.

$h/D = -0.16$ and $\theta = 90°$ are shown in Fig. 23. The flow originates from the left side of the image, the black dash circle with a light orange shade represents the propeller disk location, and the green dash line represents the PIV laser plane discussed in the previous section.

At $J = 0.1$, a distinct ring of nodal points is observed on the ceiling plane. The nodal point observed in the PIV image is denoted by yellow dots. As mentioned previously, the FOV in the PIV experiment is limited, resulting in only one of the two nodal points occurring on the PIV image plane being captured. The flow outside the propeller disk is drawn toward the propeller disk from all directions and subsequently directed downward by the propeller. The out-of-plane inflow observed in the PIV experiment occurs between these nodal points. As the advance ratio increases to 0.2, the nodal points move downstream, which is also observed in the PIV results in Fig. 19. Discontinuation of the nodal point circles is also observed closer to the trailing side of the propeller disk, indicating that the flow closer to the ceiling plane within the region is not being driven by the propeller. This also agrees with the PIV results in Fig. 19, where the reverse flow region vanished at $J \geq 0.2$.

A further increment in J results in a more pronounced disruption of the ring of nodal points due to the increment in the freestream momentum. The ring of nodal points also becomes asymmetric as it shifts further downstream on the retreating side of the propeller blade compared to the advancing side. This asymmetry is likely due to the uneven blade loading experienced by the fixed-pitch propeller

in edgewise flight, which results in a lower flow rate on the retreating side than on the advancing side. At the highest J tested, the ring of nodal points shrinks, suggesting that the interference of the ceiling plate on the propeller is diminished at higher forward flight speeds. Furthermore, the nodal point on the PIV laser plane vanishes due to the reduction in the size of the ring, and in this case, the effect of the out-of-plane inflow becomes almost negligible as it mirrors the OCE inflow condition within this region.

IV. Conclusions

The changes in propeller performance in the ceiling effect during forward flight at different incidence angles were studied experimentally. An increase in propeller thrust generation and a decrease in power consumption were observed within an h/D of -0.3 for small advance ratios ($J < 0.3$). PIV results indicated that the propeller inflow angle was increased and resembled a higher-speed edgewise flight condition due to the close proximity to the ceiling plate. With the increment of J, the ICE benefit in power and thrust reduced, as the propeller inflow pattern became closer to the OCE condition. A ring of nodal points was observed on the ceiling plane, indicating a three-dimensional flowfield in the ceiling effect. Within the nodal points, the propeller experiences an out-of-plane inflow, increasing the overall inflow angle of the propeller disk. The location of the nodal points moved downstream with an increase in J, as evidenced by phase-locked PIV and surface oil flow visualization. At higher J, the nodal points vanished, indicating a reduction of the ceiling effect on the propeller performance.

Reducing the propeller incidence angle resulted in a significant reduction in the propeller ICE performance benefit in the ceiling effect, especially at $J \geq 0.3$. Tilting the propeller nose-down in forward flight resulted in a significant reduction of the ceiling plane's effect on the propeller inflow. Hence, the propeller ceiling effect performance in hovering and low-speed forward flight cannot be used to estimate the propeller ceiling effect performance at high-speed forward flight. This brings caution to the design and mission planning for vehicles operating in the ceiling effect, where a proper balance of forward flight speed, forward tilt angle, and power saving benefit needs to be found for optimal operation.

Similar to the ceiling effect in hovering, a lower γ/D propeller has an overall higher ICE benefit in forward flight. On the other hand, a higher γ/D propeller would not benefit much from operating in close proximity to the ceiling, especially at lower incidence angles and higher forward flight speeds. In this case, the careful selection of the propeller is also essential for the design of a ceiling effect vehicle. Based on the similarity of the propeller ICE performance ($P_{\mathrm{ICE}}/P_{\mathrm{OCE}}$) in forward flight at different h/D, the overall propeller ICE performance in forward flight can be estimated utilizing both the propeller ICE performance in hovering conditions and OCE performance in forward flight.

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